

Stony Brook Institute at
Anhui University (SBIAHU)

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Applied Calculus IV

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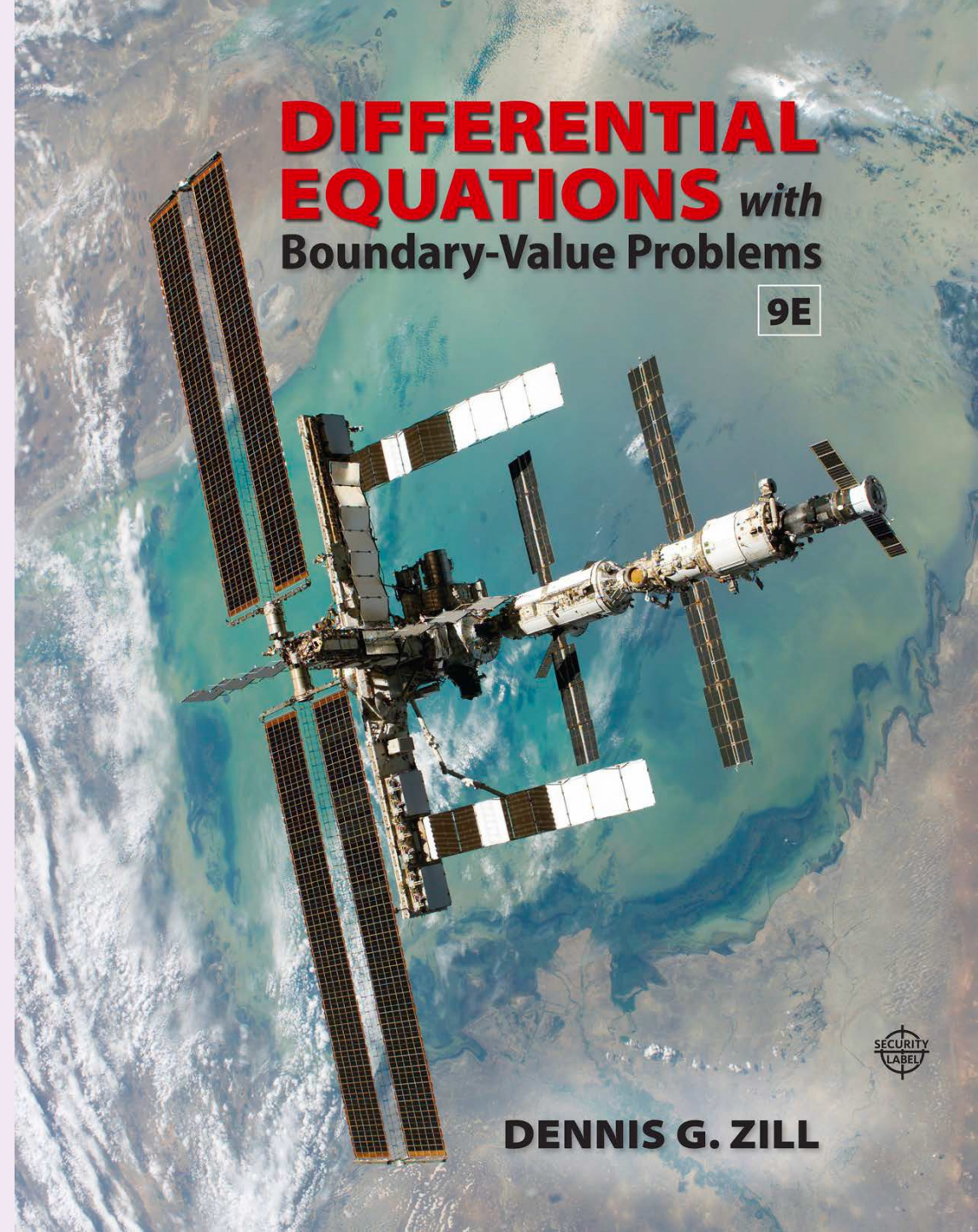
A highly recommended textbook for this course

Differential Equations

with Boundary-Value Problems

9th Edition

Dennis G. Zill



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Chapter 1



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Introduction to Differential Equations

Welcome to the study of differential equations, a powerful mathematical tool that allows us to model and understand change in the physical world. This course will equip you with the fundamental concepts, techniques, and applications essential for your STEM career.

Course Overview (Differential Equations)

What are differential equations?

Mathematical equations
containing derivatives of
unknown functions

Why are they important?

They model real-world
phenomena in physics,
engineering, biology,
economics, and more

Our approach

Starting with
definitions and
terminology,
then progress to solution
techniques & applications

- ✓ Throughout the course, we'll build our understanding from basic concepts to practical problem-solving.
- ✓ We begin with essential definitions and terminology.


$$\frac{dy}{dx} + 5y = e^x,$$

y is unknown


$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0,$$

u is unknown

What is a Differential Equation?

A differential equation is an equation containing derivatives of one or more unknown functions with respect to one or more independent variables.

$$\frac{dy}{dx} = 0.2xy.$$

independent variables



- ❖ Differential equations describe how quantities change with respect to other quantities
(they are like the codes which briefly save the changes in dynamic phenomenon)
- ❖ In this course, we'll learn to recognize, classify, and ultimately solve different types of differential equations.

Classification by Type

Ordinary Differential Equations (ODEs)

Contain only ordinary derivatives of unknown function(s) with respect to a **single** independent variable

Example: Equations describing motion of a single particle over time

Partial Differential Equations (PDEs)

Involve partial derivatives of unknown function(s) with respect to two or more independent variables

Example: Equations describing heat distribution across a surface over time

Ordinary Differential Equations (Examples)

an ODE can contain more
than one unknown function

$$\frac{dy}{dx} + 5y = e^x, \quad \frac{d^2y}{dx^2} - \frac{dy}{dx} + 6y = 0, \quad \text{and} \quad \begin{matrix} \downarrow & \downarrow \\ \frac{dx}{dt} + \frac{dy}{dt} = 2x + y \end{matrix} \quad (2)$$

Each type requires specific solution techniques that we'll explore throughout the course.

Partial Differential Equations (Examples)

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0, \quad \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2} - 2\frac{\partial u}{\partial t}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Classification by Order

The order of a differential is the order of the highest derivative in the equation.

First Order

$$4x \frac{dy}{dx} + y = x.$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Second Order

$$y'' = y' - 6y.$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0,$$

n-th Order

$$\frac{d^n y}{dx^n} = f(x, y, y', \dots, y^{(n-1)}),$$

Classification by Linearity

- Linear
- Nonlinear

Linear equations

$$(y - x) dx + 4x dy = 0, \quad y'' - 2y + y = 0, \quad x^3 \frac{d^3 y}{dx^3} + x \frac{dy}{dx} - 5y = e^x$$

Nonlinear equations

$$\begin{array}{ccc} \downarrow & & \downarrow \\ (1 - y)y' + 2y = e^x, & \frac{d^2 y}{dx^2} + \sin y = 0, & \text{and} \quad \frac{d^4 y}{dx^4} + y^2 = 0 \end{array}$$

Review (ODEs & PDEs)

Notes about ordinary differential equations (ODEs)

Single Independent Variable

Each derivative is taken with respect to only one variable (typically x or t)

Various Orders

The equations contain derivatives of different orders (first, second, etc.)

Different Forms

Linear, nonlinear, with constant or variable coefficients

Notes about Partial Differential Equations (PDEs)

Multiple Independent Variable

Partial derivatives are taken with respect to multiple independent variable

Various Orders

The equations contain partial derivatives of different orders (first, second, etc.)

Different Forms

Linear, nonlinear, with constant or variable coefficients

Also,

Real-world Applications

- Population
- Fluid dynamics
- Heat conduction across materials
- Wave propagation in fluids
- Quantum mechanics

etc.

Technically,

- Includes partial derivatives (∂) rather than ordinary derivatives
- PDEs are typically more complex to solve than ODEs and often require advanced techniques
- Often describe systems varying in space and time

Notation

Leibniz Notation

Uses fraction-like symbols to indicate derivatives:

- First derivative: dy/dx
- Second derivative: d^2y/dx^2
- n -th derivative: d^ny/dx^n

Advantage: Clearly shows the variables involved

Prime Notation

Uses prime marks to indicate derivatives:

- First derivative: y'
- Second derivative: y''
- Third derivative: y'''

Advantage: More concise in many contexts

❖ Both notations are widely used, and you'll need to be comfortable with both in this course.

Alternative Notations

Newton's Dot Notation

Commonly used for time derivatives in physics and engineering:

- First derivative: $\dot{x}$ (read as "x dot")
- Second derivative: $\ddot{x}$ (read as "x double do")

Example: $m\ddot{x} + kx = 0$ represents a simple harmonic oscillator

Subscript Notation (for Partial Derivatives)

Used primarily with partial differential equations:

- u_x means $\partial u / \partial x$
- u_{xx} means $\partial^2 u / \partial x^2$
- u_t means $\partial u / \partial t$

Example: Wave equation can be written as $u_{xx} = u_{tt}$

Notations in Practice (examples)

$$x'' + 16x = 0.$$

$$(x^2 + y^2) dx + 6xy dy = 0$$

$$\frac{dy}{dt} + 20y = 24$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$\ddot{s} = -32.$$

$$u_{xx} = u_{tt} - 2u_t$$

Differential equations use various notations that you'll encounter across different contexts and disciplines

Being fluent in translating between notations is an essential skill for advanced STEM studies

In all notations, variables, order and degree of a differential equation must be similar



DEFINITION 1.1.2 Solution of an ODE

Any function ϕ , defined on an interval I and possessing at least n derivatives that are continuous on I , which when substituted into an n th-order ordinary differential equation reduces the equation to an identity, is said to be a **solution** of the equation on the interval.

Interval of Definition

The interval I is called the interval of definition, the interval of existence, the interval of validity, or the domain of the solution.

❖ It can be an open interval (a, b) , a closed interval $[a, b]$, an infinite interval (a, ∞) , and so on.

Example: Verify that the $y = \frac{1}{16}x^4$ is a solution of $\frac{dy}{dx} = xy^{1/2}$ on the interval $(-\infty, \infty)$.

Solution

$$\text{left-hand side: } \frac{dy}{dx} = \frac{1}{16} (4 \cdot x^3) = \frac{1}{4} x^3,$$

$$\text{right-hand side: } xy^{1/2} = x \cdot \left(\frac{1}{16} x^4 \right)^{1/2} = x \cdot \left(\frac{1}{4} x^2 \right) = \frac{1}{4} x^3,$$

Question:

Is $y = \frac{1}{16}x^4$ a solution of $\frac{dy}{dx} = xy^{1/2}$ on $I = [0,1] \cup [2, \infty)$.

Solution: No

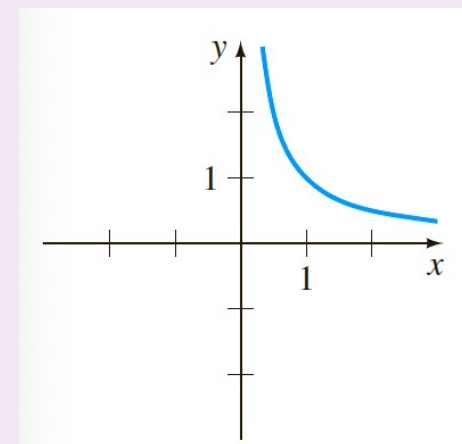
However y satisfies the differential equation, y is not continuous (not differentiable) on this domain

Example: Is $y = 1/x$ a solution of the linear first-order differential equation $x y' + y = 0$ on the interval $(0, \infty)$.

Solution:

$$x (1/x)' + 1/x = 0 \rightarrow y \text{ satisfies the differential equation}$$

y is differentiable on this domain



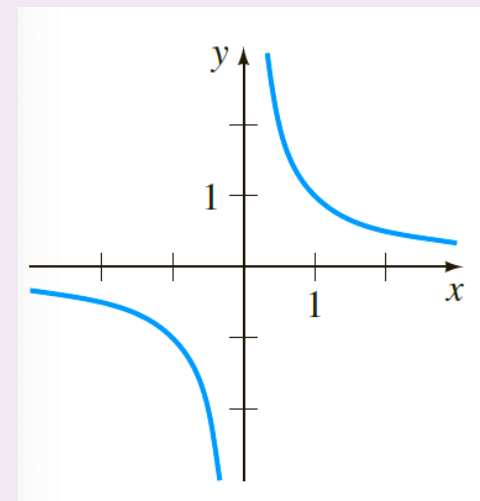
Example: Is $y = 1/x$ a solution of the linear first-order differential equation $x y' + y = 0$ on the interval $(-\infty, \infty)$.

Solution:

$$x (1/x)' + 1/x = 0 \rightarrow y \text{ satisfies the differential equation}$$

y is not differentiable on this domain

No, it is not a solution



Trivial Solution

A solution of a differential equation that is identically zero on an interval I

$$y(t) = 0$$

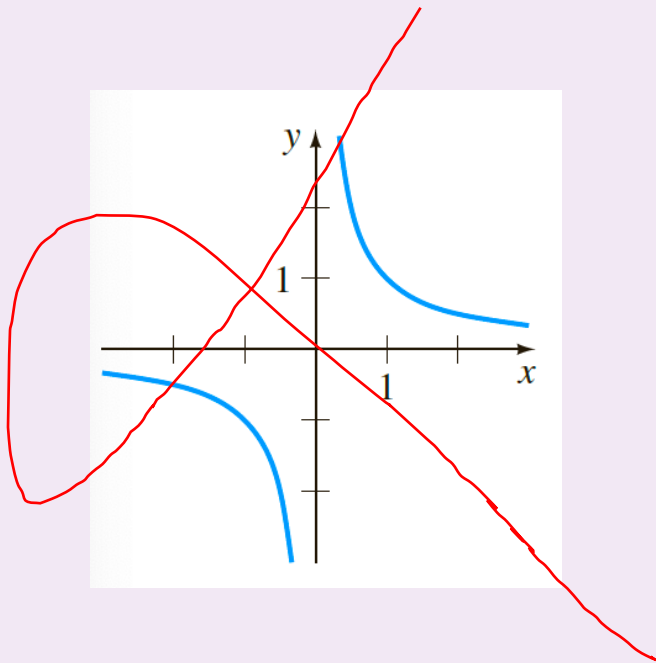
Solution Curve

The graph of a solution ϕ of an ODE is called a **solution curve**

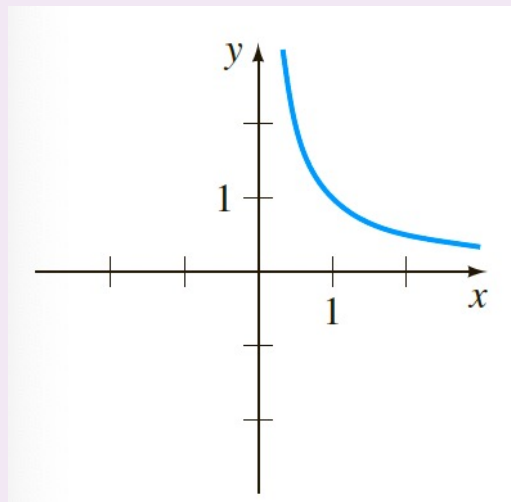
Pay attention to the interval of definition



Example: We know $y = \frac{1}{x}$ is a solution of differential equation $x y' + y = 0$ on the interval $(0, \infty)$.



Solution curve



Solutions

Explicit solution

A solution in which the dependent variable is expressed solely in terms of the independent variable

$$\frac{dy}{dx} = xy^{1/2}; \quad y = \frac{1}{16} x^4$$

Implicit solution

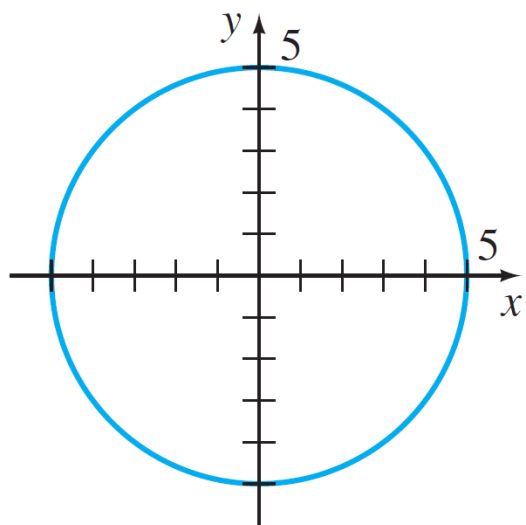
A solution which is not expressed explicitly

$$\frac{dy}{dx} = -\frac{x}{y}$$

has implicit solution $x^2 + y^2 = 25$

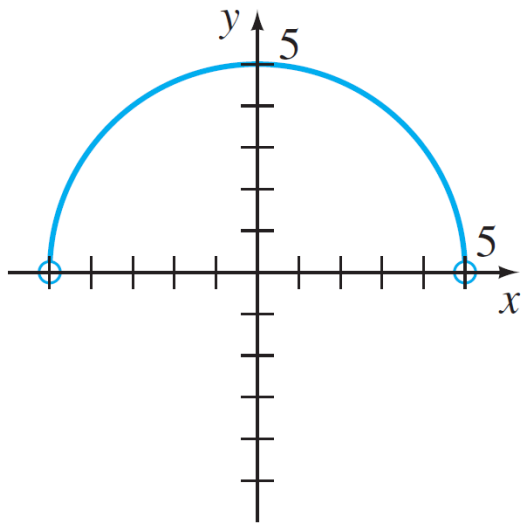
Challenge Yourself

Verify $x^2 + y^2 = 25$ is an implicit solution of $\frac{dy}{dx} = -\frac{x}{y}$ on interval $(-5, 5)$



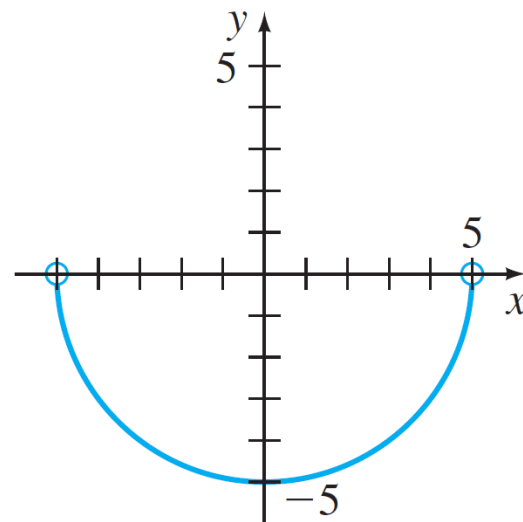
(a) implicit solution

$$x^2 + y^2 = 25$$



(b) explicit solution

$$y_1 = \sqrt{25 - x^2}, -5 < x < 5$$



(c) explicit solution

$$y_2 = -\sqrt{25 - x^2}, -5 < x < 5$$

Solutions

- Usually, there is a family of solutions which satisfy a differential equation, not a unique solution

→ Example. All the following functions

$$y(x) = x^2, \quad y(x) = x^2 + 1, \quad y(x) = x^2 + 5$$

are the solutions of differential equation $y' - 2x = 0$

- We can use parameter(s) to describe the family of solutions

→ The functions $y(x) = x^2 + c$, are the solutions of differential equation $y' - 2x = 0$

constant

Solutions

general solution

- All possible solutions which satisfy the differential equation.
- It includes parameter (how many?)

$$y' - 2x = 0, \quad y(x) = x^2 + c$$

↑
constant

particular solution

- A single solution which satisfies the differential equation
- We can find a unique solution for a differential equation if it is associated with conditions (how many?)

$$\begin{cases} y' - 2x = 0, \\ y(1) = 2 \end{cases} \quad \Rightarrow \quad y(x) = x^2 + 1$$

unique solution

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x).$$

The general solution of an *n*-th order ordinary differential equation has *n* parameters



To find a unique solution, we need *n* conditions

Example: Verify that the $y = cx - x \cos x$ is a solution of

$$xy' - y = x^2 \sin x$$

on the interval $(-\infty, \infty)$.

Example: Verify that the $y = c_1e^x + c_2xe^x$ is a solution of

$$y'' - 2y' + y = 0$$

on the interval $(-\infty, \infty)$.

Example: Verify that the $x = c_1 \cos 4t + c_2 \sin 4t$ is a solution of

$$x'' + 16x = 0$$

on the interval $(-\infty, \infty)$.

Classification by conditions

Initial-Value Problems (IVPs)

The differential equation is associated with conditions at initial points.

The value of unknown function $y(x)$ and its derivatives at a number x_0 are given.

$$\frac{dy}{dx} = xy^{1/2}, \quad y(0) = 0$$

Boundary Value Problems (BVPs)

The differential equation is associated with conditions at the end points.

The value of unknown function $y(x)$ or its derivatives are given at the end points .

$$y'' + y = 0, \quad y'(0) = 0, \quad y'(\pi) = 0$$

Examples: Initial value problems or boundary value problems?

$$2\frac{d^2y}{dx^2} - 3y^2 = 0, \quad y(0) = 0, \quad y'(0) = 1.$$

Systems of Differential Equations

A system of ordinary differential equations is two or more equations involving the derivatives of two or more unknown functions of a single independent variable.

$$\begin{cases} \frac{dx}{dt} = f(t, x, y) \\ \frac{dy}{dt} = g(t, x, y) \end{cases}$$

❖ we skip over systems of partial differential equations

Example: Verify that the pair functions

$$x = e^{-2t} + 3e^{6t},$$

$$y = -e^{-2t} + 5e^{6t}$$

is a solution of the system of differential equations

$$\frac{dx}{dt} = x + 3y$$

$$\frac{dy}{dt} = 5x + 3y;$$

on the interval $(-\infty, \infty)$.